

Part E: Dual-Path Rerouting Study Results

COL7524 – Advanced Computer Networks

1. Default route

R1–R4–R5. The path has 2 hops rather than 3 for R1–R2–R3–R5, and its configured propagation delay is also lower (4 ms versus 10 ms).

2. Routing and convergence results

Event	Route before	Route after	Convergence
Link failure at 6 s	R1–R4–R5	R1–R2–R3–R5	11.8067 ms
Link restoration at 9 s	R1–R2–R3–R5	R1–R4–R5	4.8672 ms

3. Packet delivery and loss

Interval	Transmitted	Delivered	Lost
Failure (6–9 s)	15	15	0
Recovery (9–13 s)	20	20	0
Overall (FlowMonitor)	40	40	0

4. Interpretation

- At 6 s, the active R1–R4–R5 path failed, and traffic switched to R1–R2–R3–R5. The first packet observed on the alternate path arrived 11.8067 ms after failure.
- At 9 s, the R1–R4 link was restored, and traffic returned to R1–R4–R5. The first packet observed on the restored path arrived 4.8672 ms after the restoration.
- No packet loss was observed in either interval in this run. FlowMonitor also reported 40 transmitted, 40 delivered, and 0 lost in total.

Part E: Packet Delivery and Loss by Interval

